

Muhammad Haider Ali

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Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology , Pakistan

Bachelor of Science in Computer Science

2024 – Present

Experience

Automation Engineer — Team Swift

May 2024 – Apr 2026

- Fine-tuned YOLO-based object detection models on a 500+ image UAV dataset, achieving reliable real-time aerial target detection
- Derived and implemented a geometric localization algorithm using UAV telemetry data (altitude, gimbal angle, GPS) to compute real-world target coordinates from aerial footage

Freelance Full-Stack Developer — SLMarket

2024

- Developed and deployed a full-stack e-commerce platform for a Gen Z streetwear and premium accessories brand, handling real client requirements and a production launch
- Built a role-based admin dashboard for product and inventory management with end-to-end order processing and user authentication using Node.js, Express.js, and PostgreSQL

Software Development Intern — Optome

Jun 2025 – Aug 2025

- Built a web-based customer support chatbot integrating AI-driven chat, automated FAQ resolution, and a complaint ticketing pipeline
- Designed a multi-mode response engine routing queries to AI chat, FAQ lookup, or ticket creation based on intent; integrated a JavaScript/HTML/CSS frontend with a PHP backend

Projects

GIKGO — Campus Delivery Platform

GitHub

- Built a full-stack campus delivery platform (PERN) with dual-role authentication, allowing students to switch between customer and deliverer roles within a single account
- Designed a normalized PostgreSQL schema with role-based access, location-aware deliverer availability, and a full order lifecycle (request → accept → deliver → confirm) backed by transactional integrity
- Implemented real-time availability broadcasting via WebSockets and a gamified XP-based ranking system to incentivize deliverer performance

Divido — Campus Expense Splitting App

GitHub

- Built a full-stack expense-splitting app (PERN) supporting group creation, shared expense logging, and real-time balance settlement
- Designed normalized PostgreSQL schemas and RESTful APIs with JWT-based authentication to handle multi-user group sessions securely

JWT Security Auditor — CLI Tool

GitHub

- Built a Python CLI tool that audits JWTs for algorithm confusion attacks (`alg:none`), weak secret brute-forcing via dictionary attack, missing expiry claims, and sensitive payload exposure
- Implemented HS256 secret cracking using HMAC-SHA256 against common wordlists without external cryptography dependencies

Technical Skills

Languages: C++, Python, JavaScript, PHP, RISC-V assembly

Frameworks/Libraries: Node.js, Express.js, React.js, Docker, Git

Databases: PostgreSQL

Tools: REST APIs, YOLO, Linux, WireShark, Burpsuite